

VANRA

Visual AI Network for Rights Administration

From scattered claims to explainable decisions.

Problem Statement: PS-7 - Implementation of the Forest Rights Act is hard to track across states — claims, approvals, and land-use data are fragmented, making it difficult for officials to monitor progress and flag anomalies.



VIT
BHOPAL

Origin Hackathon 2026 | **Problem Statement PS-7**

Data Science Club of VIT Bhopal

TEAM NAME: **MAVERICK TRIO**

TEAM MEMBERS' NAMES: **UMANG PATEL, VARUN VAISH, MUNNA BABU ANSARI**

▶ LIVE DEMO [<https://origin-red.vercel.app/>]



GITHUB
[<https://github.com/umang25mei10037-coder/ORIGIN>]

The Problem

FRA implementation is fragmented across systems and slowing down justice.

4.5M+

forest rights
claims across India

Contextual scale

Records are **fragmented** across Gram Sabha, SDLC, and District Administration
Processing delays often exceed the **180-day guideline**

Forest boundaries and revenue land records may **not align**

Officials lack a **unified prioritization** system

Data exists — but **actionable intelligence is missing**.



Fragmented
Data

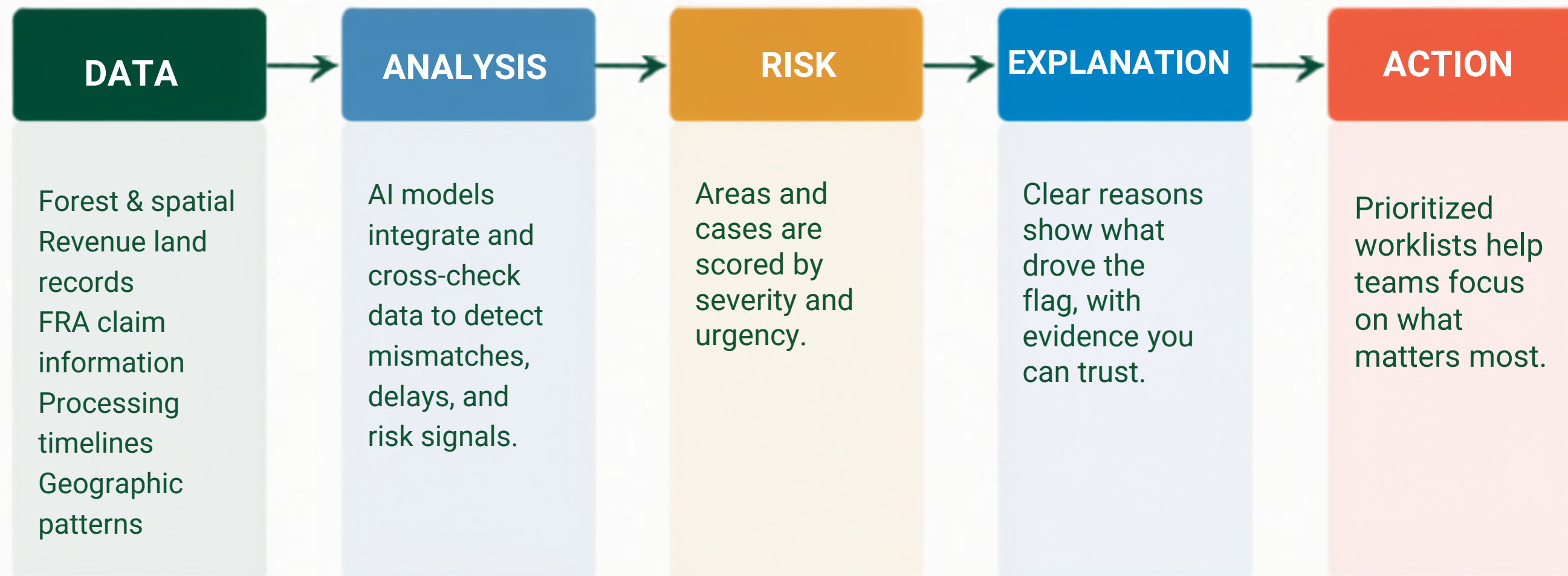
Processing
Delays

Boundary & Record
Mismatches

Difficult
Prioritization

Our Solution — VANRA.

VANRA brings together forest, land and claim data, turns it into explainable insights, and guides action.



Key questions VANRA answers

WHERE is the problem?

WHY was it flagged?

HOW URGENT is it?

WHAT should be investigated first?

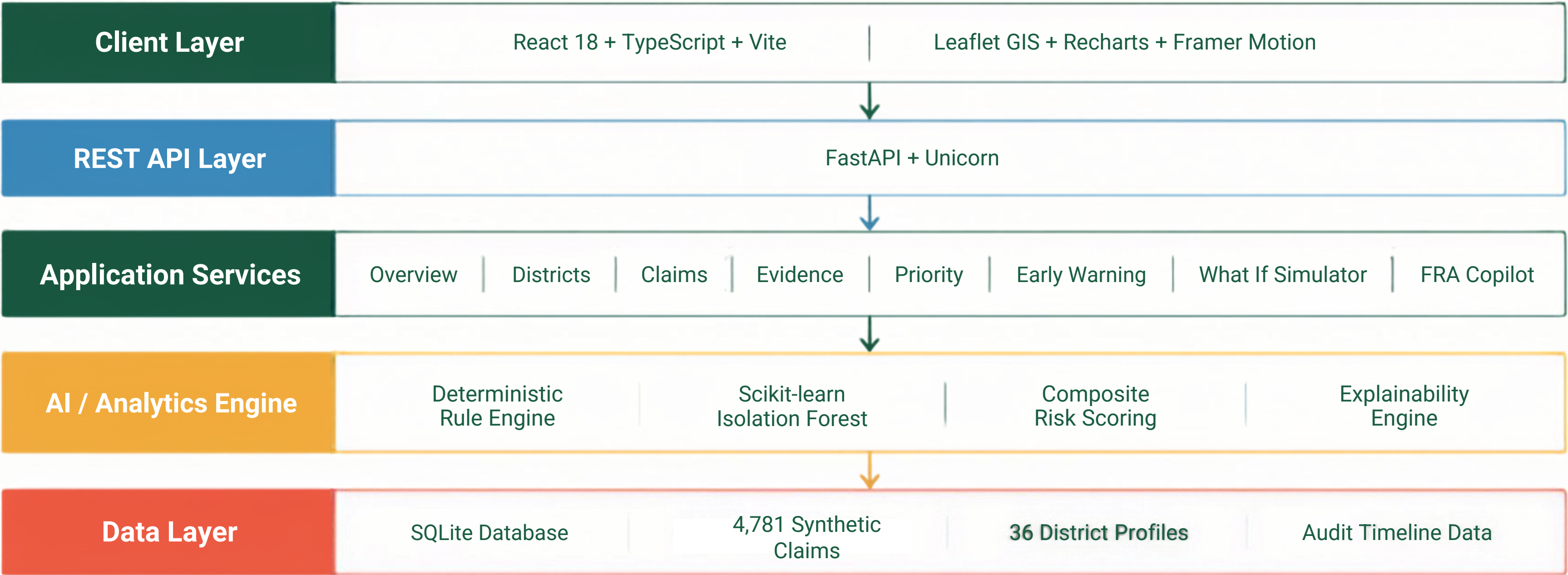
VANRA does not grant or reject forest rights.

It supports authorized decision-makers with explainable, actionable insights.

VANRA Workflow.



System Architecture.



Offline-capable | Modular | Auditable

SYNTHETIC DEMONSTRATION DATA

Key Features.

DEMO LINKS:

- **GIS MAP:** <https://origin-red.vercel.app/overview>
- **CLAIM:** <https://origin-red.vercel.app/claims>
- **EARLY WARNING:** <https://origin-red.vercel.app/early-warning>
- **AI Priority Queue:** <https://origin-red.vercel.app/priority>
- **Simulator:** <https://origin-red.vercel.app/simulator>

01

Living GIS Map

Claims, risk levels,
district health,
spatial clusters

02

District Intelligence

High-risk districts
and score factors

03

Claim Investigation

statutory workflow &
processing timeline

04

Evidence Graph

Claims linked to land
records, parcels, forest
zones, workflow stages

05

AI Priority Queue

Immediate, High,
Medium priority
ranking

06

Early Warning

approaching delays,
backlogs, emerging
anomalies

07

What If Simulator

simulate effect of
increasing
processing capacity

08

FRA Copilot

structured Analysis
→
Recommendation
→ Data Basis



Unified view.
From claims
to decisions.



Faster & fairer.
Data-led prioritization,
human-led decisions.

Stronger forests.
Better governance,
better outcomes.

How Our AI Works.

FRA COPILOT DEMO LINK - <https://origin-red.vercel.app/copilot>

Hybrid AI engine with explainable, decomposable risk scoring.

Deterministic Rule Engine

Checks known risk signals with clear, consistent rules

Processing delay >180 days

Record mismatch

Area discrepancy

Workflow bottleneck

Isolation Forest

Multidimensional outlier detection

Flags unusual combinations that don't follow the norm

Composite Risk Score

78

/100

Every risk score is explainable and decomposable.

Why this case is flagged

Top factor contribution (example)

Processing Delay

32%

Record Mismatch

27%

Geographic Cluster

19%

Area Anomaly

12%

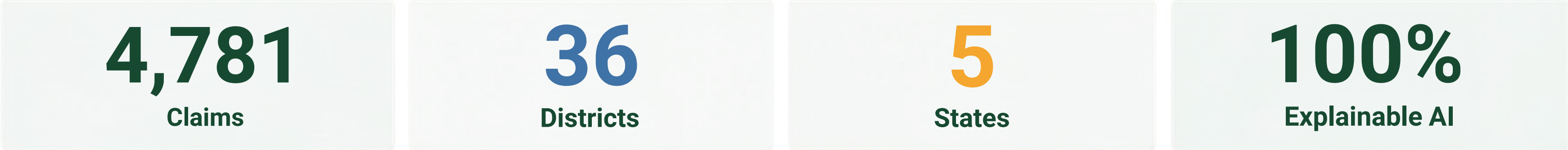
Workflow Bottleneck

10%

AI identifies what deserves attention. Humans make the final decision.

Why VANRA Matters.

Actionable insights today. Responsible scaling tomorrow.



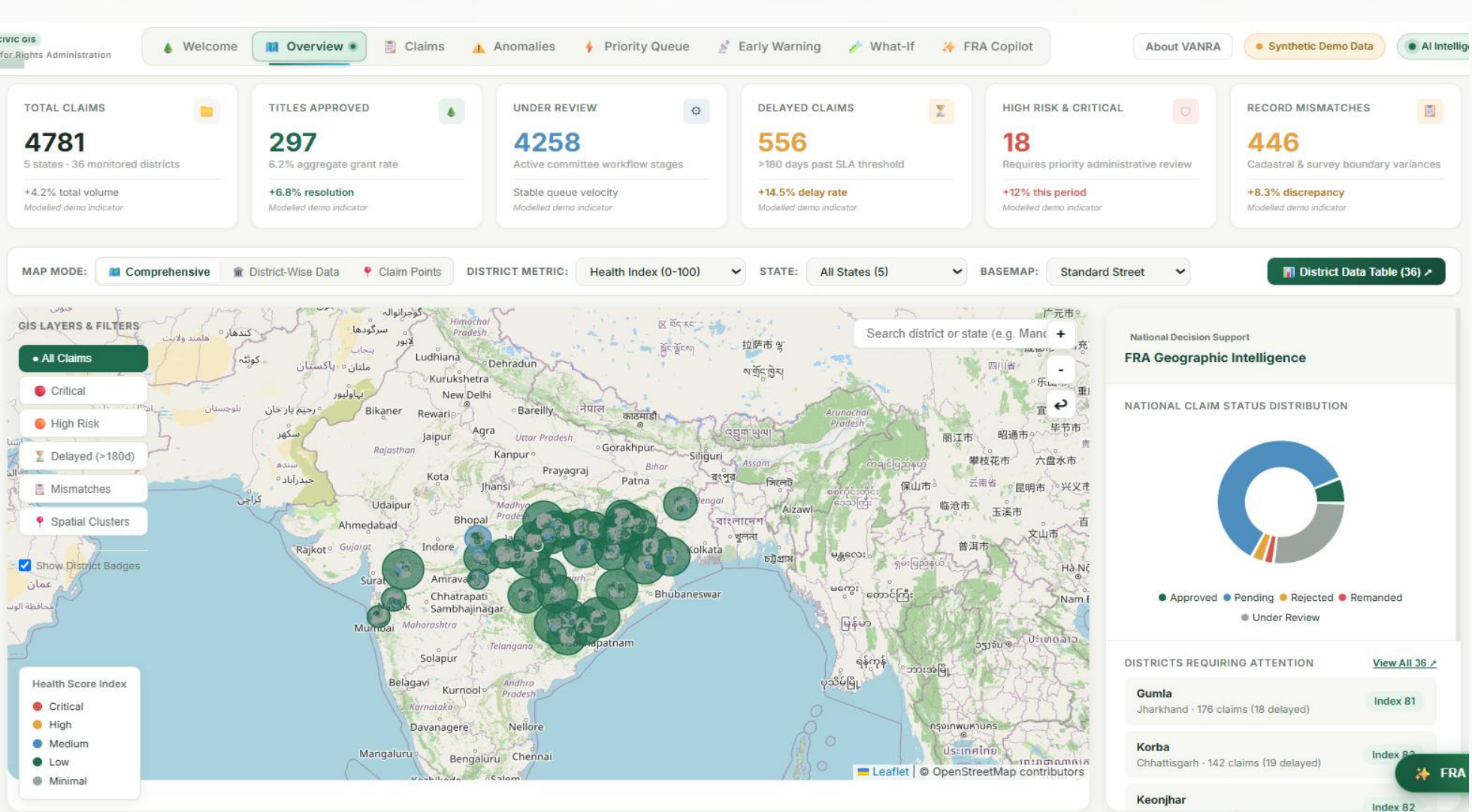
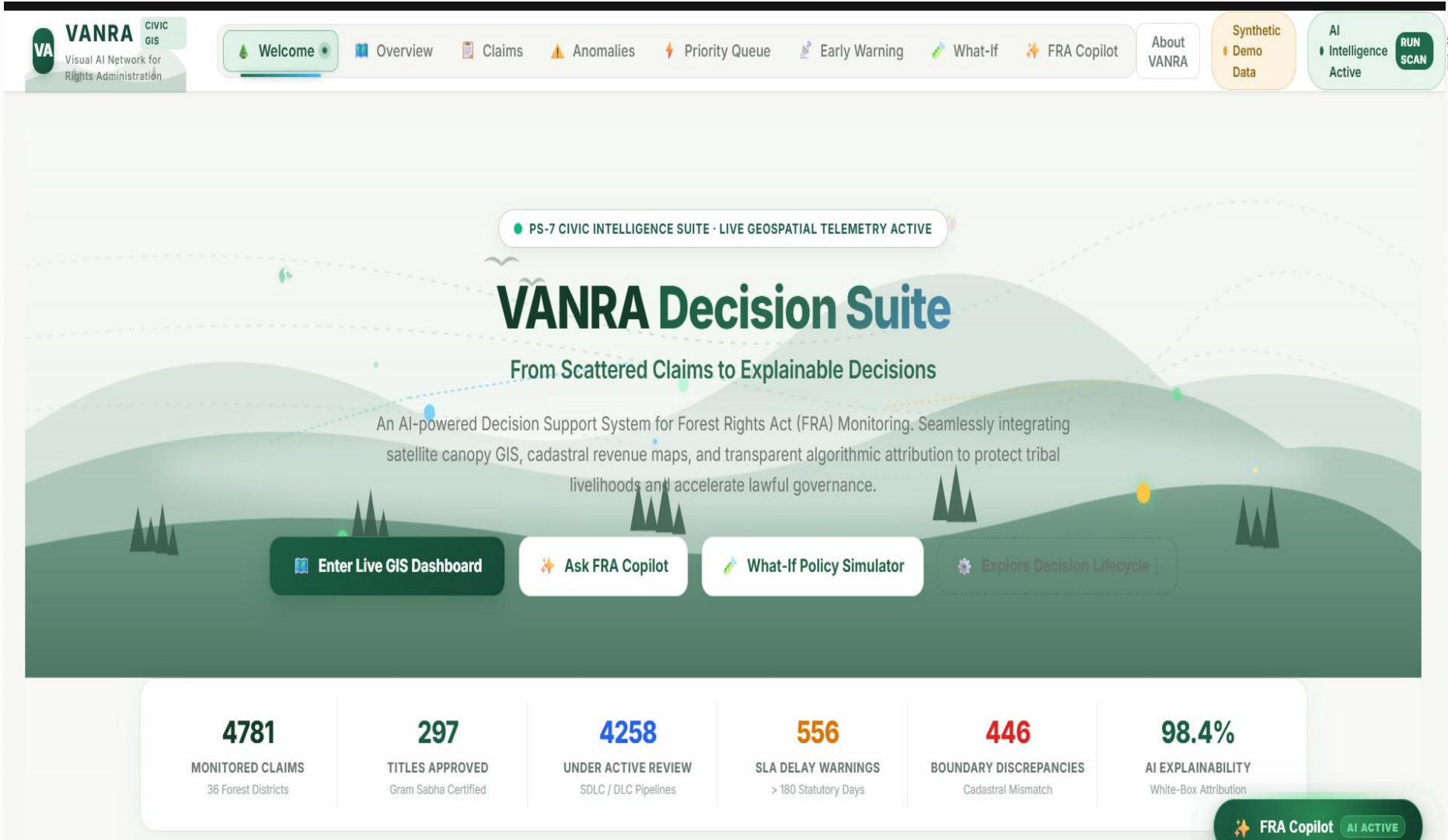
Synthetic demonstration data

Impact Highlights	Trust & Safety	Roadmap & Security
Faster identification of delayed claims Geographic visibility of emerging problems Evidence-based prioritization Transparent AI reasoning Reduced manual investigation effort	Human-in-the-loop no automated claim rejection Synthetic demo data no real citizen PII Administrative language only Offline-capable architecture	Prototype → Pilot → Secure Deployment RBAC Encryption Audit Trails DPDPA Alignment

VANRA is AI-powered decision support. It does not grant or reject forest rights — AI identifies what deserves attention, humans make the final decision.

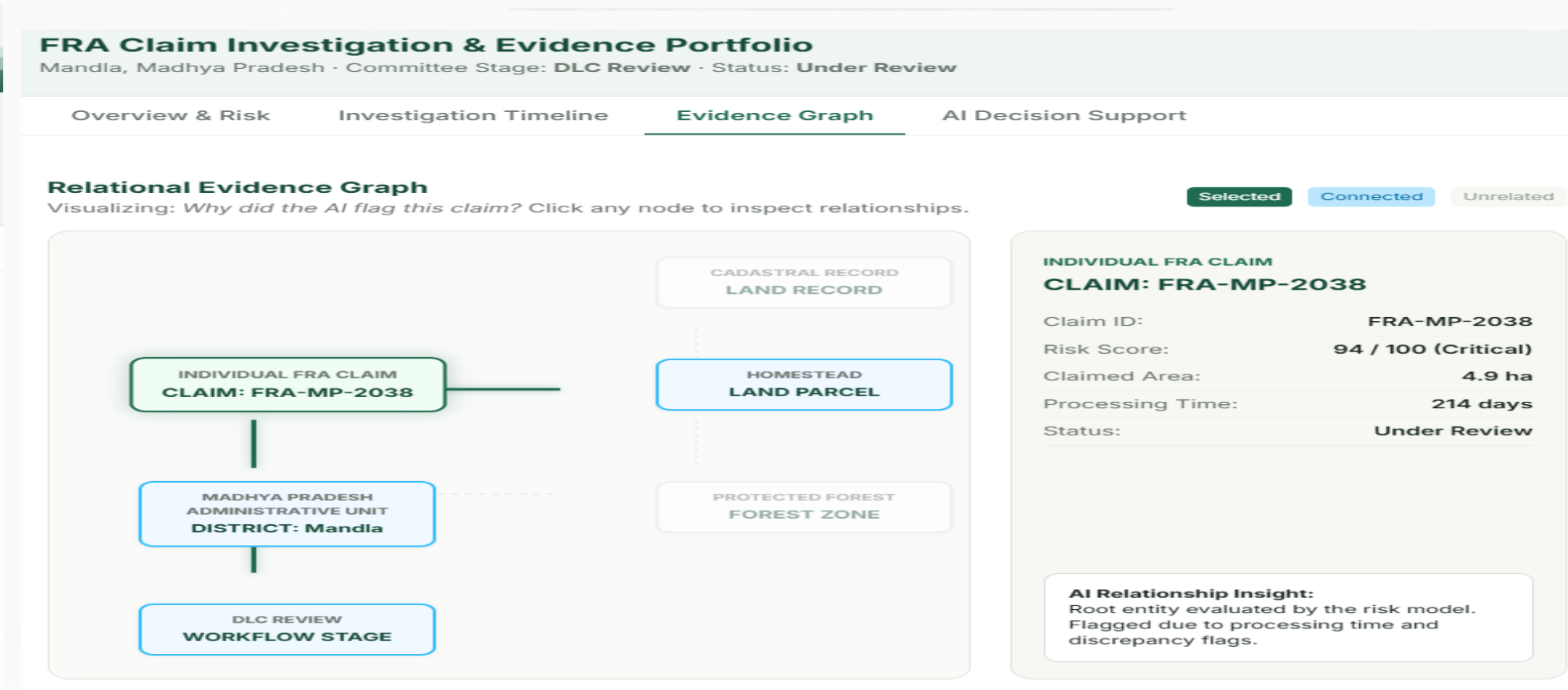
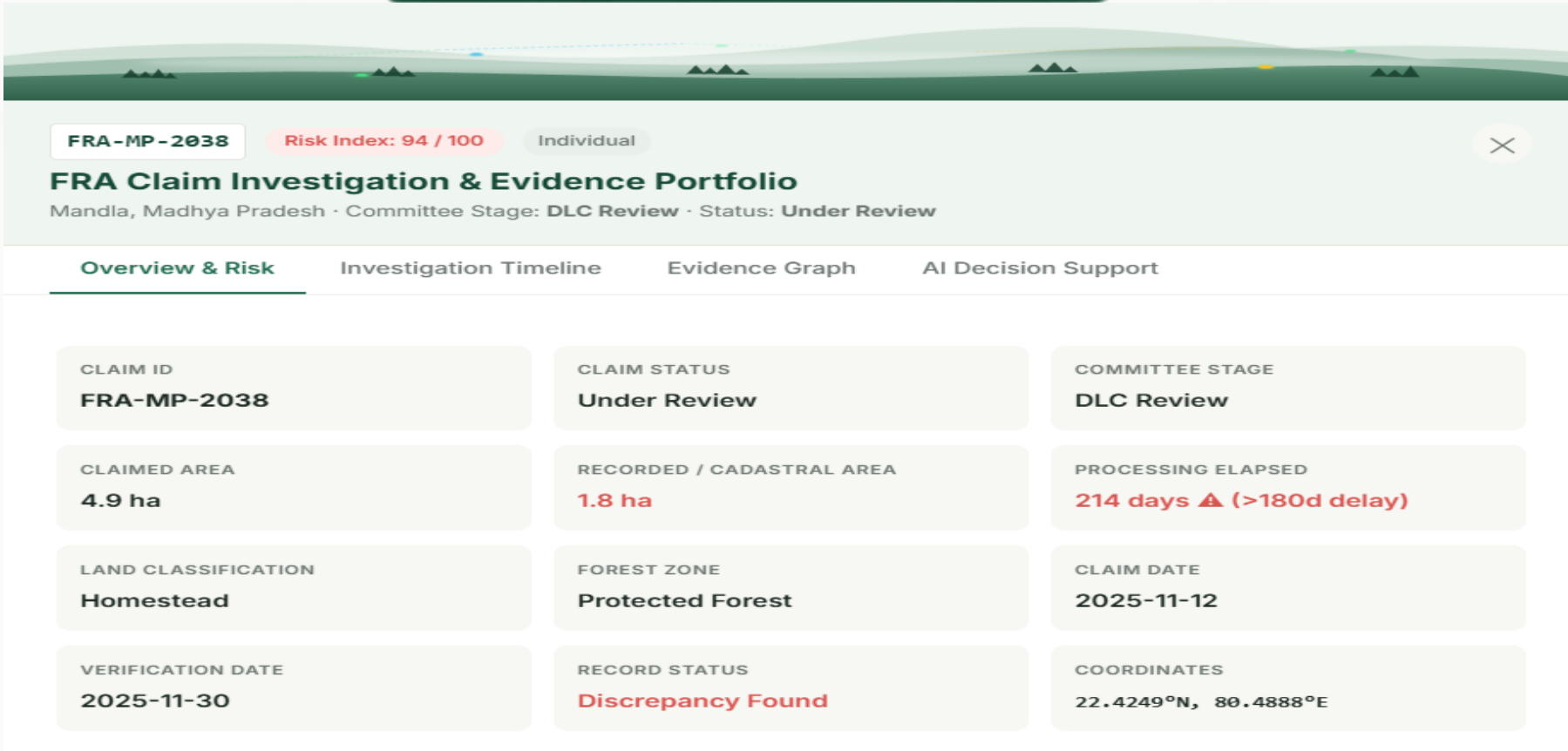
VANRA DASHBOARD

VANRA GIS MAP

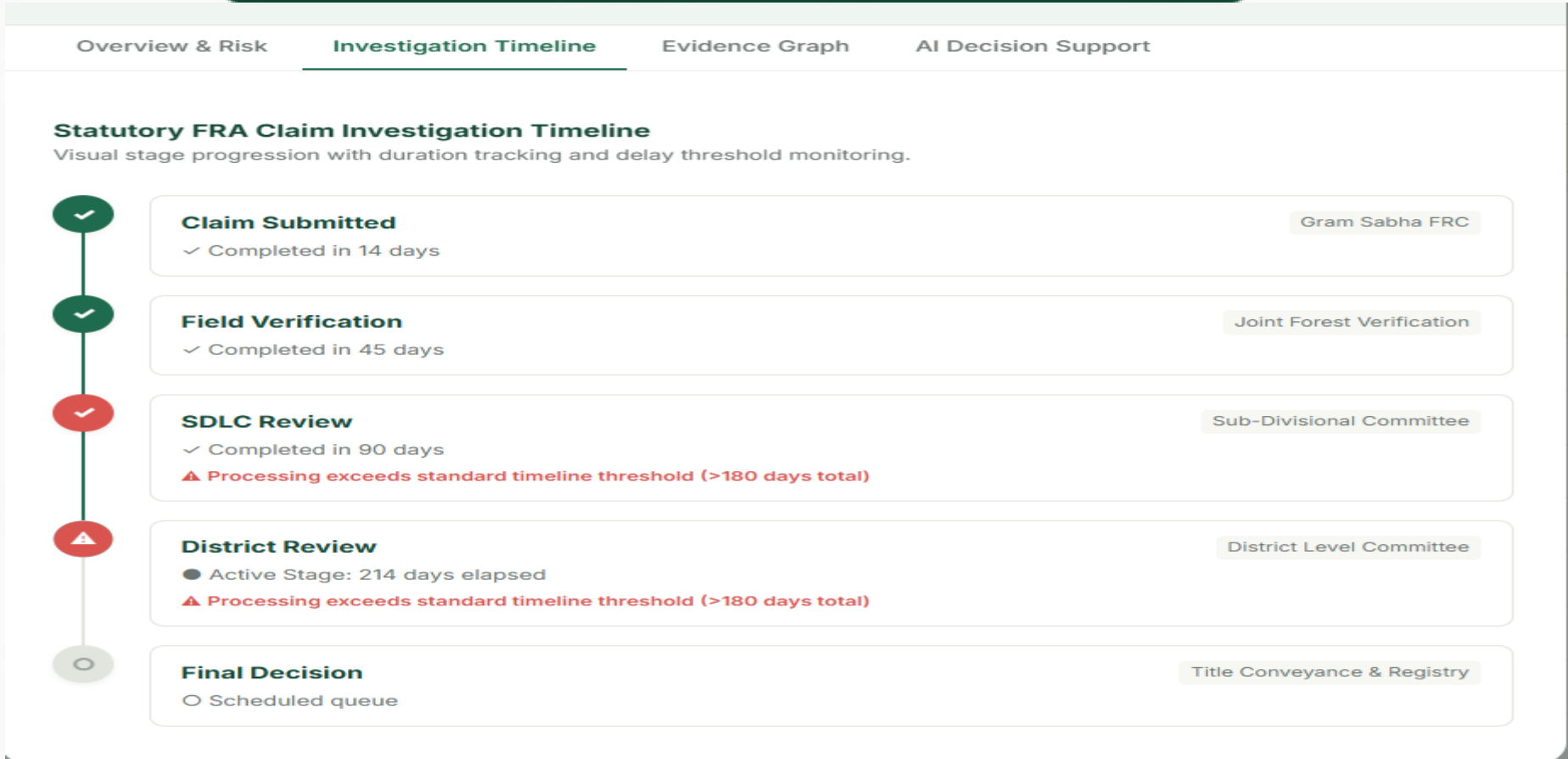


CLAIM INVESTIGATION

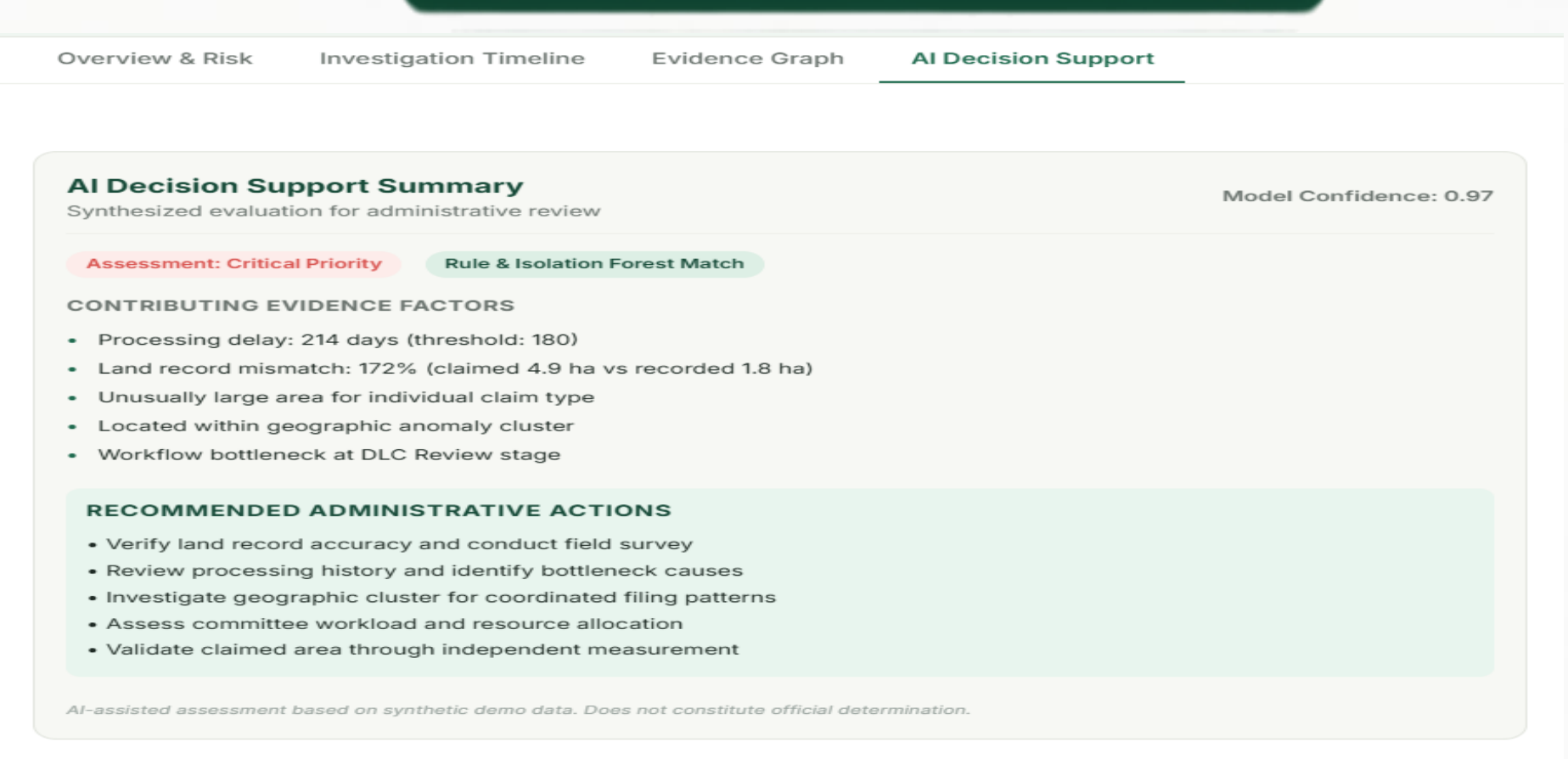
EVIDENCE GRAPH



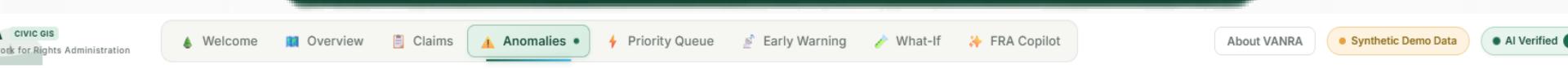
INVESTIGATION TIME LINE



AI DECISION SUPPORT



Anomaly Detection Engine



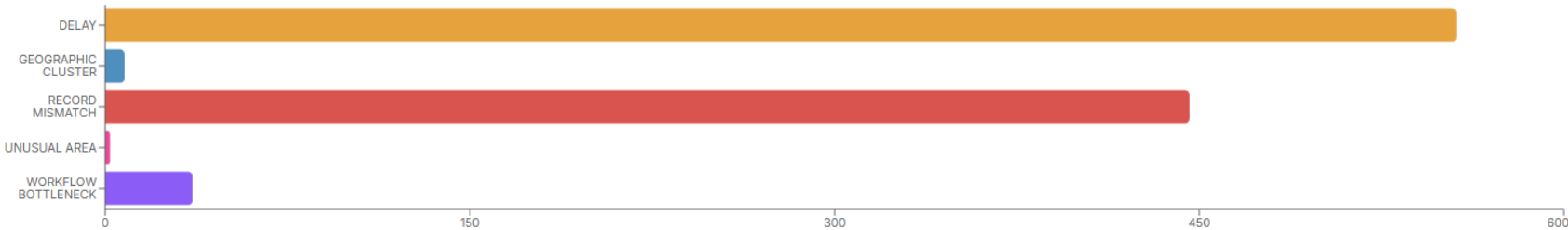
Anomaly Detection Engine

Deterministic business rules + Isolation Forest anomaly detection flags across claim stages
100 anomaly occurrences

▲ Algorithmic detection based on synthetic demo data. Flags indicate cases for review, not legal infractions.

Anomaly Category Distribution

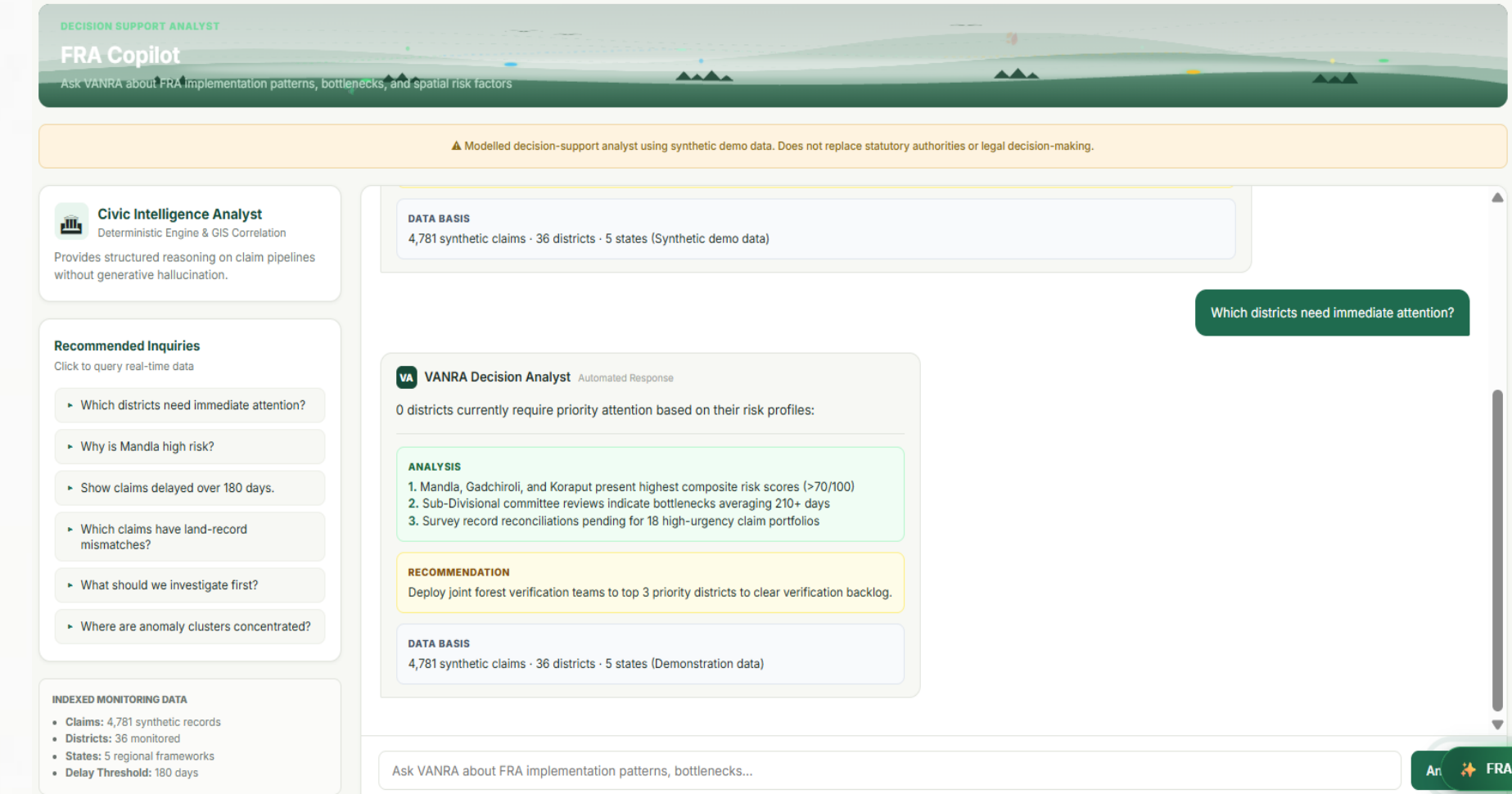
Relative frequencies of detected administrative signals



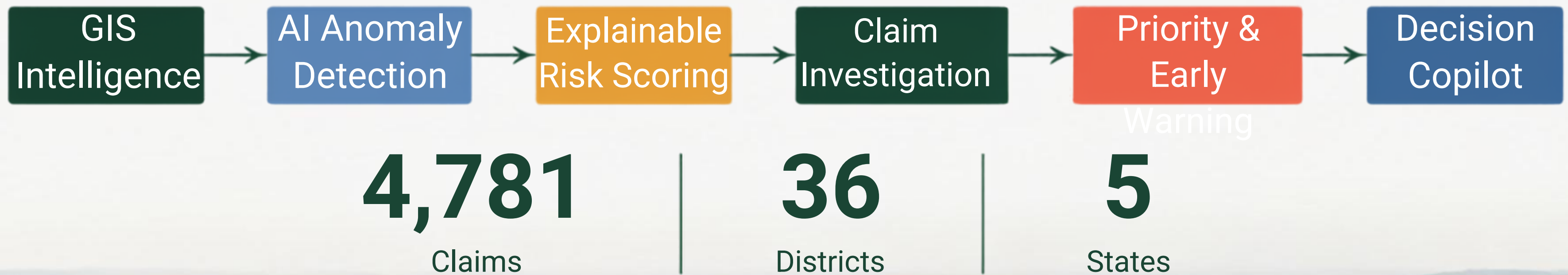
All Anomalies DELAY RECORD MISMATCH GEOGRAPHIC CLUSTER WORKFLOW BOTTLENECK UNUSUAL AREA

FRA-MP-2038	RECORD MISMATCH	Mandla, Madhya Pradesh — Area discrepancy: 172% (claimed 4.9 ha vs recorded 1.8 ha)	Critical	+25 pts
FRA-MP-0028	RECORD MISMATCH	Mandla, Madhya Pradesh — Area discrepancy: 157% (claimed 2.52 ha vs recorded 0.98 ha)	Critical	+25 pts

FRA COPILOT



VANRA – From Scattered Claims to Explainable Decisions



LIVE VANRA

[<https://origin-red.vercel.app/>]

VANRA does not grant or deny forest rights.
It gives decision-makers the **intelligence**
to know where to look first.